

## Problem

Design a problem to help other students better understand magnitude and frequency scaling.

## Step-by-step solution

## Step 1 of 1

Design a problem to help other students better understand magnitude and frequency scaling.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

**Problem**

What values of  $K_m$  and  $K_f$  will scale a 4-mH inductor and a 20- $\mu$ F capacitor to 1 H and 2 F respectively?

**Solution**

$$L'C' = \frac{LC}{K_f^2} \longrightarrow K_f^2 = \frac{LC}{L'C'}$$

$$K_f^2 = \frac{(4 \times 10^{-3})(20 \times 10^{-6})}{(1)(2)} = 4 \times 10^{-8}$$

$$K_f = \underline{2 \times 10^{-4}}$$

$$\frac{L'}{C'} = \frac{L}{C} K_m^2 \longrightarrow K_m^2 = \frac{L'}{C'} \cdot \frac{C}{L}$$

$$K_m^2 = \frac{(1)(20 \times 10^{-6})}{(2)(4 \times 10^{-3})} = 2.5 \times 10^{-3}$$

$$K_m = \underline{5 \times 10^{-2}}$$